KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 28 · READING

Wings Over Michoacán — Teacher Lesson Plan & Answer Key

Learning objectives: Students start from a research question, take notes in their own words that name their sources, decide which facts are relevant, tell copying, paraphrasing and synthesizing apart, combine facts from an article, a fact sheet and a data table into synthesis sentences that cite each source, and report a range with both citations when reliable sources disagree.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name capture: Naveen and Nayeli are research partners in a homeschool science club in Lawrence, Kansas, where Monarch Watch has asked volunteers to tag monarch butterflies since 1992. Their research question: Why do eastern monarchs fly to the mountain forests of Michoacán, Mexico, and how are they doing today? The student types first and last name so results go to KingCretot Experience. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, and the answer plus the reason appear after the 2nd miss. Typed one-word answers ignore capital letters and are accepted when the answer contains an accepted word.
2. Slide 1 — Warm-Up: Week 27 Review (8 pts, 4 each): red herrings, alibis, first-person telling and inference.
3. Slide 2 — Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade4-5/prepost/KCE_Reading_PrePost_G4-5W28.pdf) and completes Page 1 (Pre-Lesson Warm-Up) on paper before any teaching.
4. Slide 3 — Picture It: Putting Sources Together. A research question, notes in your own words that each name a source, crossing out what is not relevant, then synthesizing and citing. Model question: “Why is the library garden full of bees this summer?” Copy (exact words in quotation marks), paraphrase (one source), synthesize (two or more sources).
5. Slide 4 — Vocabulary Round-Up: 8 flip cards (research question, source, paraphrase, synthesize, cite, relevant, overwinter, cluster).
6. Slide 5 — Your Turn: Using the Words (8 pts, 4 each): one multiple-choice and one typed vocabulary item.
7. Slide 6 — Source A: Winter in the Fir Forests (5 pts). A journey of up to about 3,000 miles to high forests in Michoacán and the State of Mexico; arrival around Día de Muertos (November 1 and 2); oyamel fir forests from about October to March, with clusters on the trunks; the Monarch Butterfly Biosphere Reserve named a UNESCO World Heritage Site in 2008; and the colony Catalina Aguado and Kenneth Brugger found on Cerro Pelón on January 2, 1975.
8. Slide 7 — Source B: The Super Generation (5 pts). Most adults live about a month; late-summer monarchs go into diapause and save energy; the super generation lives around eight months and flies 50 to 100 miles a day; in March females lay eggs only on milkweed; three to five generations finish the trip north.
9. Slide 8 — Source C: The Data, and Your Note Cards (10 pts, 5 each): the acres table (about 45 acres in 1996 to 1997; 2.22, 4.42 and 7.24 acres from 2023 to 2026), better weather credited for the jump, threats (milkweed loss, drought, illegal logging), and the December 2024 proposal to list the monarch as threatened. Items check the table, which notes are relevant, and citations.

10. Slide 9 — Teach: Writing a Synthesis Sentence. Sort, connect, combine, cite, with a chart of the best note for each part of the question. When sources disagree ("around eight months" versus a United States Fish and Wildlife Service "8 to 9 months"), report a range and cite both.
11. Slide 10 — Synthesis Detective (12 pts, 6 each): the best synthesis, why researchers cite, what is relevant, a disagreement between sources, and citation mistakes.
12. Slide 11 — Fix the Sentence, Part 1 (8 pts, 4 each): each item names the one kind of fix. The whole sentence is checked exactly (straight or curly quotes and extra spaces are OK). The corrected sentence appears after the 2nd miss. Michoacán and Día may be typed with or without the accent mark.
13. Slide 12 — Fix the Sentence, Part 2 (8 pts, 4 each): two more sentences, same rules (a comma before but, subject-verb agreement, fewer or less, who's or whose, than or then, and a past-tense verb).
14. Slide 13 — Back It Up With Evidence (4 pts): quote the source sentence named in the item, starting with the tag shown (such as Source A states), a comma, and the exact words in quotation marks with the end mark inside. Checked exactly.
15. Slide 14 — Writer's Turn: Synthesize It (6 pts): one typed synthesis auto-checked by rules: a minimum word count; no 4 or more words in a row copied from the original (allowed names count as one word); and every key idea included, including the names of the sources. Feedback names the rule that was missed, and a model answer appears after the 2nd miss. Encourage covering the original before writing.
16. Slide 15 — Match-Up Round (10 pts): research words matched to meanings, facts matched to the source they came from, or words from the sources matched to meanings (a different set each attempt), 2 pts each on the first try.
17. Slide 16 — Speed Trail (bonus XP only): 3 quick research questions against the student's own best time.
18. Slide 17 — Boss Challenge (16 pts, 8 each): two-step items across all three sources (decide what the question needs, then check the evidence and the citations).
19. Slide 18 — Pencil Break (Post): the student completes Page 2 (Post-Lesson Activity) of the same sheet (grade4-5/prepost/KCE_Reading_PrePost_G4-5W28.pdf), including the ★ Challenge writing task if time allows, before seeing the score.
20. Slide 19 — Results (Research Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows "Master Researcher!" with the congratulations message "Master researcher! You put the sources together!" and a big confetti burst; below 85 shows "Keep Going!" with encouragement ("You're getting closer, and every try makes you stronger."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: WEEK 27 REVIEW (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
`rv1_s0`	Set 1: Nanami's library book is missing. Her brother Nasir walks by holding a book with a blue cover, just like hers, but it turns out to be his own. Later Nanami spots a blue corner poking out from under the couch cushions. Which detail is the RED HERRING? [Nasir holding a blue book that turns out to be his own / The blue corner under the couch cushions / Nanami noticing that her book is missing / The couch in the living room]	**Nasir holding a blue book that turns out to be his own**	Nasir's blue book points toward him, but it is his own book, so it points the wrong way. The quiet clue is the blue corner under the cushions (Week 27).

ID	Question	Answer	Why (explain to student)
rv2_s0	Set 1: Type your answer: when the cookies vanished, Nasir was at swim practice, and his coach saw him the whole time. In a mystery, proof like this that a suspect could not have done it is called an ___.	**alibi** (also accepted with "a" or "an" or in the plural)	Proof that a suspect was somewhere else is an alibi. Mrs. Mitra's point about Moxie's short leash worked like an alibi (Week 27).
rv1_s1	Set 2: A mystery begins: "I knew the missing key had to be somewhere in the barn." Who is telling this mystery? [A character in the story, using "I" (first person) / A narrator outside the story (third person) / Nobody; mysteries do not have narrators / The reader of the story]	**A character in the story, using "I" (first person)**	The word "I" shows a character is telling the story in the first person, like Montrell in "The Case of the Vanishing Hand Pies" (Week 27).
rv2_s1	Set 2: Type your answer: a decision you reach by putting the clues together with what you already know is an ___.	**inference** (also accepted: inferring; with "a" or "an" or in the plural)	An inference is a smart conclusion built from clues plus what you know. Mitsuko inferred that someone with a key opened the case (Week 27).
rv1_s2	Set 3: Find the mistake! Neela writes, "The red herring in a mystery is always the real answer." Which explanation is correct? [A red herring points the wrong way; the quiet, real clues lead to the answer. / Neela is right; the loudest clue is always the answer. / A red herring is the detective's alibi. / A red herring is the theme of the mystery.]	**A red herring points the wrong way; the quiet, real clues lead to the answer.**	Red herrings are planted to mislead. In both Savannah mysteries, the loud clue was wrong and a quiet clue solved the case (Week 27).
rv2_s2	Set 3: Type your answer: in last week's two mysteries, the first character to be blamed turned out to be innocent each time. A character who might be responsible for the problem is called a ___.	**suspect** (also accepted with "a" or "an" or in the plural)	A character who might have done it is a suspect. The man with the backpack and Moxie the beagle were both suspects who turned out to be innocent (Week 27).

YOUR TURN: USING THE WORDS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: From Source A: "The butterflies overwinter, from about October to March, in forests of oyamel fir trees." What does overwinter most likely mean? [To spend the winter in one place / To fly over the winter snow / To win a winter race / To freeze solid for the winter]	**To spend the winter in one place**	The butterflies stay in the forests from about October to March, the winter months, so overwinter means to spend the winter in one place.
v2_s0	Set 1: Type your answer: to combine ideas from two or more sources into one new statement is to ___.	**synthesize** (also accepted: synthesizing, synthesis)	Combining ideas from several sources is synthesizing. Paraphrasing uses only one source.
v1_s1	Set 2: From Source A: "The monarchs pack together in close clusters on the trunks and branches." What is a cluster? [A group of things packed close together / A single butterfly resting alone / A tall fir tree / A long flight south]	**A group of things packed close together**	"Pack together" and "close" are context clues: a cluster is a group packed close together.
v2_s1	Set 2: Type your answer: when Nayeli writes "(Source B)" after a fact, she names where she found it. Naming your source this way is called ___ it. Type one word.	**citing** (also accepted: cite, cited)	To cite a source is to name where a fact came from. (Careful: "site" means a place, and "sight" means seeing.)

ID	Question	Answer	Why (explain to student)
v1_s2	Set 3: Which explanation of the difference between paraphrasing and synthesizing is correct? [Paraphrasing retells one source in your own words; synthesizing combines ideas from two or more sources. / They mean exactly the same thing. / Synthesizing means copying a source word for word. / Paraphrasing means using five or more sources at once.]	**Paraphrasing retells one source in your own words; synthesizing combines ideas from two or more sources.**	Both use your own words, but synthesizing joins ideas from more than one source into a new statement.
v2_s2	Set 3: Type your answer: a fact that helps answer your research question is ___. A fact that does not help, like the library's opening time, should be crossed out.	**relevant**	A relevant fact connects to your question. Researchers cross out facts that are interesting but not relevant.

SOURCE A: THE ARTICLE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
sa_s0	Set 1: Type your answer: according to Source A, paragraph 3, the monarchs spend the winter in forests of what kind of tree? Type the two-word answer.	**oyamel fir** (also accepted: oyamel; with "a" or "an" or in the plural)	Paragraph 3: "The butterflies overwinter, from about October to March, in forests of oyamel fir trees."
sa_s1	Set 2: Source A, paragraph 2, says many monarchs arrive around Día de Muertos. According to the paragraph, what do the returning monarchs mean to many people? [A symbol of the souls of loved ones coming back to visit / A sign that summer is starting / A warning that a storm is coming / A reason to cut down the fir trees]	**A symbol of the souls of loved ones coming back to visit**	Paragraph 2: "To many people, the returning monarchs are a symbol of the souls of loved ones coming back to visit."
sa_s2	Set 3: Type your answer: according to Source A, paragraph 4, in what year was the Monarch Butterfly Biosphere Reserve named a World Heritage Site?	**2008**	Paragraph 4: "In 2008, the Monarch Butterfly Biosphere Reserve ... was named a World Heritage Site."

SOURCE B: THE FACT SHEET (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
sb_s0	Set 1: According to Source B, paragraph 2, what puts late-summer monarchs into diapause? [Shorter days and cooler temperatures / Eating too much milkweed / Flying 100 miles in one day / Arriving in the fir forests]	**Shorter days and cooler temperatures**	Paragraph 2: "Shorter days and cooler temperatures put them into diapause."
sb_s1	Set 2: Type your answer: according to Source B, paragraph 4, on what plant do female monarchs lay their eggs?	**milkweed** (also accepted with "a" or "an" or in the plural)	Paragraph 4: "Females lay their eggs only on milkweed, the one plant monarch caterpillars depend on."
sb_s2	Set 3: According to Source B, paragraph 3, about how long does a member of the super generation live? [Around eight months / About one month / About eight days / About three years]	**Around eight months**	Paragraph 3: "They live about eight times as long as summer monarchs, around eight months." Paragraph 1 says summer monarchs live only about a month.

SOURCE C: THE DATA, AND YOUR NOTE CARDS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
qp1_s0	Set 1: Using Source C, which statement about the 2024 to 2025 winter is correct? [The colonies covered 4.42 acres, about double the 2.22 acres of the winter before. / The colonies covered 7.24 acres, the most ever recorded. / The colonies covered fewer acres than the winter before. / The colonies covered about 45 acres, the same as in 1996 to 1997.]	**The colonies covered 4.42 acres, about double the 2.22 acres of the winter before.**	2.22 × 2 = 4.44, so 4.42 acres is about double. The 1996 to 1997 record, about 45 acres, is much larger than any recent winter.
qp2_s0	Set 1: Type your answer: two steps. Nayeli's note card says "about 45 acres, the most ever recorded." Find that row in Source C. In which winter did the colonies cover that much forest? Type the FIRST year of that winter.	**1996**	The table shows about 45 acres in the 1996 to 1997 winter, the most ever recorded, so her card should say 1996 to 1997.
qp1_s1	Set 2: Which note card is RELEVANT to the second part of the question, "How are the monarchs doing today?" [Source C: illegal logging still damages forests in the reserve. / Source A: the reserve is about 60 miles from Mexico City. / Source B: super generation monarchs can fly 50 to 100 miles a day. / Source A: Día de Muertos is celebrated on November 1 and 2.]	**Source C: illegal logging still damages forests in the reserve.**	Logging is a threat to the monarchs today, so it helps answer "How are they doing?" The other facts are true but answer different questions.
qp2_s1	Set 2: Type your answer: two steps. In Source C, find the winter with the biggest jump, then read the first note. Scientists credited the increase to better ___. Type one word.	**weather**	The first note says, "Scientists credited better weather, with less drought, for the big increase in the 2024 to 2025 winter."
qp1_s2	Set 3: Find the mistake! Nayeli's note card says: "Source B: the colonies covered 7.24 acres in the 2025 to 2026 winter." What is wrong? [The fact is correct, but it comes from Source C, so the card names the wrong source. / The number should be 72.4 acres. / Nothing is wrong with the card. / The fact is about Día de Muertos, not acres.]	**The fact is correct, but it comes from Source C, so the card names the wrong source.**	Source B is the fact sheet about the super generation. The acres are in Source C's table. A correct fact with the wrong source is still a citation mistake.
qp2_s2	Set 3: Type your answer: two steps. In Source C, find the acres for the 2025 to 2026 winter and for the 1996 to 1997 winter. Did the colonies cover MORE or FEWER acres in 2025 to 2026? Type more or fewer.	**fewer** (also accepted: less)	7.24 acres is far less than about 45 acres, so the colonies covered fewer acres in 2025 to 2026, even after growing for two winters.

SYNTHESIS DETECTIVE (6 POINTS EACH)

ID	Question	Answer	Why (explain to student)
d1_s0	Set 1: Which sentence best SYNTHESIZES Sources A and B to explain why monarchs fly to Mexico? [Source B explains that late-summer monarchs stop breeding and save energy for a long flight, and Source A shows where it ends: cool, steady fir forests in central Mexico. / Source A says the reserve became a World Heritage Site in 2008. / These forests give them a cool, steady climate for the winter. / Source B says monarchs spend the winter in Canada, and Source A agrees.]	**Source B explains that late-summer monarchs stop breeding and save energy for a long flight, and Source A shows where it ends: cool, steady fir forests in central Mexico.**	The best answer joins a fact from each source and names both. One choice uses only one source, one copies Source A without quotation marks or a citation, and one gets the facts wrong.
d2_s0	Set 1: Which explanation of WHY researchers cite their sources is correct? [Citing tells readers where each fact came from so they can check it, and it gives credit to the people who found it. / Citing makes a report longer, which earns a better grade. / Citing is only needed when a fact is false. / Citing means you do not have to use your own words.]	**Citing tells readers where each fact came from so they can check it, and it gives credit to the people who found it.**	Citations let readers check facts and give credit. Even with citations, notes and reports still need your own words, unless you use quotation marks.
d1_s1	Set 2: Nayeli wants one sentence that combines Sources B and C. Which is best? [Monarch caterpillars depend on milkweed (Source B), so the loss of milkweed in the United States, one threat listed in Source C, puts future monarchs at risk. / Monarchs can fly 50 to 100 miles a day (Source B). / Monarch caterpillars eat oyamel fir needles (Source B), so logging is a threat (Source C). / The colonies covered 7.24 acres (Source A), and milkweed matters (Source C).]	**Monarch caterpillars depend on milkweed (Source B), so the loss of milkweed in the United States, one threat listed in Source C, puts future monarchs at risk.**	The best answer connects a fact from Source B to a threat in Source C and cites each correctly. One choice uses one source, one invents a fact about fir needles, and one cites the wrong sources.
d2_s1	Set 2: Source B says the super generation lives “around eight months.” A United States Fish and Wildlife Service fact sheet says the migrating generation “can live 8 to 9 months.” What should a careful researcher do? [Report a range, such as about eight to nine months, and cite both sources. / Pick the bigger number and cite neither source. / Decide that both sources are useless. / Write that scientists proved the answer is exactly 8.5 months.]	**Report a range, such as about eight to nine months, and cite both sources.**	Both sources are reliable and close. Reporting a range with both citations is honest. Averaging them and calling it “proven” would invent a fact.
d1_s2	Set 3: Find the mistake! Naveen writes, “According to Source A, monarchs can fly 50 to 100 miles a day.” Which explanation is correct? [The fact is in Source B, paragraph 3, not Source A, so the citation is wrong. / Naveen is right; the fact is in Source A, paragraph 1. / The fact is false; monarchs fly only 5 miles a day. / The fact is in Source C’s table.]	**The fact is in Source B, paragraph 3, not Source A, so the citation is wrong.**	Source B, paragraph 3: “They can fly 50 to 100 miles a day.” Source A talks about the forests and the reserve, not daily flying distance.

ID	Question	Answer	Why (explain to student)
d2_s2	Set 3: Which fact is NOT relevant to the question “How are the monarchs doing today?” [Catalina Aguado and Kenneth Brugger found a colony on Cerro Pelón in 1975. / The colonies covered 7.24 acres in the 2025 to 2026 winter. / Illegal logging damages forests in the reserve. / In December 2024, the United States Fish and Wildlife Service proposed listing the monarch as threatened.]	**Catalina Aguado and Kenneth Brugger found a colony on Cerro Pelón in 1975.**	The 1975 discovery is important history, but it does not tell how the monarchs are doing now. The other three facts describe their numbers and threats today.

FIX THE SENTENCE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Introductory phrase: add the missing comma. According to Source C the colonies covered 7.24 acres.	**According to Source C, the colonies covered 7.24 acres.**	Put a comma after an introductory phrase like “According to Source C.”
fs2_s0	Set 1: Capital letters: fix the capital-letter mistakes in the three place names. Monarchs fly from east of the rocky mountains to michoacán, mexico.	**Monarchs fly from east of the Rocky Mountains to Michoacán, Mexico.** (also accepted: Monarchs fly from east of the Rocky Mountains to Michoacan, Mexico.)	Names of specific places are proper nouns: Rocky Mountains, Michoacán, Mexico.
fs3_s0	Set 1: Compound sentence: add the comma before the joining word “but.” The colonies grew last winter but they are still smaller than in the 1990s.	**The colonies grew last winter, but they are still smaller than in the 1990s.**	When “but” joins two complete sentences, put a comma before it.
fs4_s0	Set 1: Grammar: fix the verb so it agrees with the subject. The forests of Michoacán gives the butterflies a cool climate.	**The forests of Michoacán give the butterflies a cool climate.** (also accepted: The forests of Michoacan give the butterflies a cool climate.)	The subject is “forests” (plural), not “Michoacán,” so the verb is “give.”
fs1_s1	Set 2: Article title: put quotation marks around the title of the article. The article Winter in the Fir Forests explains where monarchs rest.	**The article "Winter in the Fir Forests" explains where monarchs rest.**	The title of an article goes inside quotation marks.
fs2_s1	Set 2: Commas in a series: add the commas between the three sources. Naveen read an article a fact sheet and a data table.	**Naveen read an article, a fact sheet, and a data table.** (also accepted: Naveen read an article, a fact sheet and a data table.)	Use commas to separate three items in a series: an article, a fact sheet, and a data table.
fs3_s1	Set 2: Fewer or less: fix the word. Use “fewer” for things you can count. There were less butterflies in the forest that winter.	**There were fewer butterflies in the forest that winter.**	Butterflies can be counted, so use “fewer.” Use “less” for amounts you cannot count, like less water.
fs4_s1	Set 2: Grammar: fix the who’s / whose mistake. Catalina Aguado, who's hike made history, found a colony in 1975.	**Catalina Aguado, whose hike made history, found a colony in 1975.**	“Who’s” means “who is.” The hike belongs to Catalina Aguado, so use the possessive “whose.”
fs1_s2	Set 3: American punctuation: move the period to the right spot. Source B says monarchs "can fly 50 to 100 miles a day".	**Source B says monarchs "can fly 50 to 100 miles a day."**	In American English, a period at the end of a quotation goes inside the closing quotation mark.

ID	Question	Answer	Why (explain to student)
fs2_s2	Set 3: Capital letters: fix the holiday names. Many monarchs arrive around día de muertos, the day of the dead.	**Many monarchs arrive around Día de Muertos, the Day of the Dead.** (also accepted: Many monarchs arrive around Dia de Muertos, the Day of the Dead.)	Holidays are proper nouns, so capitalize the important words: Día de Muertos, the Day of the Dead.
fs3_s2	Set 3: Than or then: fix the word. The super generation lives longer then its parents.	**The super generation lives longer than its parents.**	Use “than” to compare. “Then” tells when something happens.
fs4_s2	Set 3: Grammar: fix the verb so it matches the past time “In 1975.” In 1975, Catalina Aguado and Kenneth Brugger find a colony.	**In 1975, Catalina Aguado and Kenneth Brugger found a colony.**	The discovery happened in the past, so use the past-tense verb “found.”

BACK IT UP WITH EVIDENCE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
qt_s0	Set 1: This sentence explains WHY the fir forests are a good winter home. Quote it. Original (Source A, paragraph 3): These forests give them a cool, steady climate for the winter. Start with: Source A states	**Source A states, "These forests give them a cool, steady climate for the winter."**	Put a comma after the tag, then copy every exact word, including the comma after “cool,” inside quotation marks, with the period inside.
qt_s1	Set 2: This sentence explains why milkweed matters so much. Quote it. Original (Source B, paragraph 4): Females lay their eggs only on milkweed, the one plant monarch caterpillars depend on. Start with: Source B states	**Source B states, "Females lay their eggs only on milkweed, the one plant monarch caterpillars depend on."**	Put a comma after the tag, then copy every exact word, including the comma after “milkweed,” inside quotation marks, with the period inside.
qt_s2	Set 3: This sentence tells what the government has done about the monarch. Quote it. Original (Source C, third note): In December 2024, the United States Fish and Wildlife Service proposed listing the monarch as a threatened species. Start with: Source C reports	**Source C reports, "In December 2024, the United States Fish and Wildlife Service proposed listing the monarch as a threatened species."**	Put a comma after the tag, then copy every exact word, including the comma after “2024,” inside quotation marks, with the period inside.

WRITER'S TURN: SYNTHESIZE IT (6 POINTS EACH)

ID	Question	Answer	Why (explain to student)
wt_s0	Set 1: In one or two sentences, explain WHY eastern monarchs fly to Mexico's mountain forests. Use a fact from Source A AND a fact from Source B, and name both sources. Source A, paragraph 3: The butterflies overwinter, from about October to March, in forests of oyamel fir trees. These forests give them a cool, steady climate for the winter. The monarchs pack together in close clusters on the trunks and branches, so many that some branches bend under their weight. Scientists think clustering helps the butterflies hold in heat. Source B, paragraph 2: The monarchs that hatch in late summer, in August or September, are different. Shorter days and cooler temperatures put them into diapause, a kind of pause in which they do not mate or lay eggs. Instead, they save their energy for the long flight south. Use at least 20 words. Do not copy 4 or more words in a row from the sources. (Names like Michoacán are fine.) Name Source A (the article). Name Source B (the fact sheet). Give a fact about the forests: the fir trees or the cool, steady climate. Give a fact about the butterflies: diapause, saving energy, or not laying eggs. Original: "Every fall, millions of monarch butterflies that hatched east of the Rocky Mountains, in the United States and Canada, begin a journey of up to about 3,000 miles. They are heading for a few high mountain forests in central Mexico, in the ..." (the full passage is on screen; it is long, so only its opening is shown here)	**Sample:** Source B explains that monarchs born in late summer stop breeding and store up energy for a huge trip, and Source A says they rest all winter in chilly, calm fir woods high in Mexico.	A strong answer names Source A and Source B, gives a fact about the fir forests or their cool climate, and a fact about diapause or saving energy, all in new words. Auto-check rubric: at least 20 words; no 4 or more words in a row copied from the original (names count as one word); must include: source a or article; source b or fact sheet; fir or forest or tree or cool or climate or steady or mountain; diapause or energy or egg or breed or mate or pause or super or rest.

ID	Question	Answer	Why (explain to student)
wt_s1	Set 2: In one or two sentences, combine Source B and Source C: explain why milkweed matters and how the monarchs are doing today. Name both sources. Source B, paragraph 4: In March, the super generation leaves Mexico and flies north. Females lay their eggs only on milkweed, the one plant monarch caterpillars depend on. It takes three to five generations of monarchs to finish the trip back to the northern United States and Canada. Source C: 2.22 acres (2023 to 2024), 4.42 acres (2024 to 2025), 7.24 acres (2025 to 2026), about 45 acres in 1996 to 1997 (the most ever recorded). Threats to monarchs include the loss of milkweed and flowering plants in the United States, caused by changes in how land is used and by weed killers and insect killers; drought and extreme weather; and illegal logging in the reserve's forests. Use at least 20 words. Do not copy 4 or more words in a row from the sources. Name Source B (the fact sheet). Name Source C (the data). Mention milkweed. Tell how they are doing: the acres, growing, smaller, or a threat. Original: "Most adult monarch butterflies live only about a month. They begin laying eggs when they are just a few days old. The monarchs that hatch in late summer, in August or September, are different. Shorter days and cooler temperatures put them …" (the full passage is on screen; it is long, so only its opening is shown here)	**Sample:** Source B says monarch caterpillars need milkweed to survive, and Source C shows the colonies have grown for two winters but remain far smaller than in the 1990s, partly threatened by vanishing milkweed.	A strong answer names Source B and Source C, explains that caterpillars depend on milkweed, and tells how the monarchs are doing (growing but still far below the 1996 to 1997 record, with threats), in new words. Auto-check rubric: at least 20 words; no 4 or more words in a row copied from the original (names count as one word); must include: source b or fact sheet; source c or data or survey or table; milkweed; acre or smaller or fewer or grew or growing or increase or rose or declin or drop or less or threat or risk or danger.
wt_s2	Set 3: In one or two sentences, answer the WHOLE research question: why eastern monarchs go to Michoacán AND how they are doing today. Cite at least two sources by name. The research question: Why do eastern monarchs fly to the mountain forests of Michoacán, Mexico, and how are they doing today? Use your notes from Sources A, B, and C. Use at least 22 words. Do not copy 4 or more words in a row from the sources. Tell WHY they go: the cool fir forests, the winter, or diapause. Tell HOW they are doing: the acres, growth, or a threat. Cite your sources by name, like Source A or Source C. Original: "Every fall, millions of monarch butterflies that hatched east of the Rocky Mountains, in the United States and Canada, begin a journey of up to about 3,000 miles. They are heading for a few high mountain forests in central Mexico, in the …" (the full passage is on screen; it is long, so only its opening is shown here)	**Sample:** The super generation saves its energy and spends the winter in chilly fir woods high in central Mexico (Sources A and B), and while the colonies have grown since 2024, they are still much smaller than in the 1990s and face logging and drought (Source C).	A strong answer tells why they go (cool fir forests for the winter, diapause), how they are doing (growing but far below the 1996 to 1997 record, with threats), and cites at least two sources, in new words. Auto-check rubric: at least 22 words; no 4 or more words in a row copied from the original (names count as one word); must include: fir or forest or tree or cool or climate or winter or mountain or diapause or energy; acre or smaller or fewer or grew or growing or increase or declin or threat or risk or logging or milkweed or drought; source.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match source	**where information comes from**	2 points if matched on the first try.
match_s0_2	Set 1: Match cite	**name where a fact came from**	2 points if matched on the first try.
match_s0_3	Set 1: Match paraphrase	**retell one source in your own words**	2 points if matched on the first try.
match_s0_4	Set 1: Match synthesize	**combine ideas from several sources**	2 points if matched on the first try.
match_s0_5	Set 1: Match relevant	**helps answer the question**	2 points if matched on the first try.
match_s1_1	Set 2: Match oyamel fir forests	**Source A: the winter home**	2 points if matched on the first try.
match_s1_2	Set 2: Match diapause	**Source B: a pause in laying eggs**	2 points if matched on the first try.
match_s1_3	Set 2: Match 7.24 acres	**Source C: the 2025 to 2026 colonies**	2 points if matched on the first try.
match_s1_4	Set 2: Match Cerro Pelón, 1975	**Source A: a famous discovery**	2 points if matched on the first try.
match_s1_5	Set 2: Match milkweed	**Source B: where eggs are laid**	2 points if matched on the first try.
match_s2_1	Set 3: Match overwinter	**spend the winter in one place**	2 points if matched on the first try.
match_s2_2	Set 3: Match cluster	**a group packed close together**	2 points if matched on the first try.
match_s2_3	Set 3: Match super generation	**late-summer monarchs that migrate**	2 points if matched on the first try.
match_s2_4	Set 3: Match threatened species	**a living thing at risk of becoming endangered**	2 points if matched on the first try.
match_s2_5	Set 3: Match research question	**the question a report answers**	2 points if matched on the first try.

BOSS CHALLENGE (8 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: Two-step: Step 1, which part of the research question does Source C answer? Step 2, which note proves it? Which answer gets BOTH steps right? [How the monarchs are doing today; note: the colonies covered 7.24 acres in 2025 to 2026, far below about 45 acres in 1996 to 1997. / How the monarchs are doing today; note: the super generation lives around eight months. / Why the monarchs go to Mexico; note: the colonies covered 7.24 acres in 2025 to 2026. / Where the monarchs go; note: Catalina Aguado found a colony in 1975.]	**How the monarchs are doing today; note: the colonies covered 7.24 acres in 2025 to 2026, far below about 45 acres in 1996 to 1997.**	Source C is the survey data, so it answers “How are they doing?” The acres are the proof. The lifespan fact is from Source B, and the 1975 discovery is from Source A.

ID	Question	Answer	Why (explain to student)
b2_s0	Set 1: Two-step: Naveen claims, “The super generation is built for its long winter away.” Step 1, choose evidence from TWO sources. Step 2, check the citations. Which answer gets BOTH steps right? [It lives around eight months instead of about one (Source B), long enough to spend October to March in Mexico (Source A). / It lives around eight months (Source C), long enough to spend the winter in Canada (Source A). / It lives around eight months instead of about one (Source B). / Many people see the monarchs as souls of loved ones (Source A), and they fly 100 miles a day (Source C).]	**It lives around eight months instead of about one (Source B), long enough to spend October to March in Mexico (Source A).**	The best answer combines a lifespan fact from Source B with the October-to-March stay from Source A, and cites both correctly. One choice has wrong facts and citations, one uses just one source, and one is not about lifespan at all.
b1_s1	Set 2: Two-step: the colonies grew from 2.22 acres to 4.42 acres to 7.24 acres. Step 1, have the monarchs fully recovered? Step 2, which evidence shows it? [Not yet: 7.24 acres is still far below the roughly 45 acres of 1996 to 1997, and Source C lists threats that remain. / Yes: the colonies doubled once, so the monarchs are fully recovered. / Yes: the government made a final decision to protect them in 2024. / Not yet: the reserve became a World Heritage Site in 2008.]	**Not yet: 7.24 acres is still far below the roughly 45 acres of 1996 to 1997, and Source C lists threats that remain.**	Growth is good news, but 7.24 acres is much less than about 45. Source C says the December 2024 listing was only a proposal, with no final decision yet, and the 2008 honor does not measure the population.
b2_s1	Set 2: Which NEW research situation shows synthesis done CORRECTLY? [Nevin reads a bird guide and a park ranger’s blog, notes which said what, and writes one sentence in his own words that combines both, naming each source. / Nevin copies a whole paragraph from a bird guide and adds the name of the website at the end. / Nevin reads three sources but writes about only the one he likes best, and never names it. / Nevin reads one source and lists three source names so his report looks longer.]	**Nevin reads a bird guide and a park ranger’s blog, notes which said what, and writes one sentence in his own words that combines both, naming each source.**	Synthesis means combining ideas from more than one source, in your own words, with each source named. Copying, ignoring sources, and fake citations are not synthesis.
b1_s2	Set 3: Find the mistake! Nadira writes, “Monarchs fly up to 3,000 miles to Mexico, where they lay their eggs on oyamel fir trees (Source A).” Which explanation is correct? [Source A never says that: Source B explains that the super generation does not lay eggs during diapause and that females lay eggs only on milkweed after leaving Mexico in March. / Nadira is right; Source A says eggs go on fir trees. / The only mistake is the distance; it should be 300 miles. / The only mistake is that she should have cited Source C.]	**Source A never says that: Source B explains that the super generation does not lay eggs during diapause and that females lay eggs only on milkweed after leaving Mexico in March.**	The 3,000 miles is from Source A, but the egg-laying part is wrong. Source B says the super generation is in diapause (no eggs) and lays eggs on milkweed when it flies north.

ID	Question	Answer	Why (explain to student)
b2_s2	Set 3: Two-step: Step 1, choose the best answer to the WHOLE research question. Step 2, check that every citation is correct. Which answer gets BOTH steps right? [Late-summer monarchs pause breeding and fly to cool fir forests in central Mexico for the winter (Sources A and B); their colonies grew to 7.24 acres in 2025 to 2026, but threats like milkweed loss and logging remain (Source C). / Late-summer monarchs pause breeding and fly to cool fir forests (Source C); their colonies grew to 7.24 acres (Source B). / Monarchs go to Mexico to celebrate Día de Muertos (Source A), and they are fully recovered (Source C). / Monarchs fly 50 to 100 miles a day (Source B).]	**Late-summer monarchs pause breeding and fly to cool fir forests in central Mexico for the winter (Sources A and B); their colonies grew to 7.24 acres in 2025 to 2026, but threats like milkweed loss and logging remain (Source C).**	The best answer covers both parts of the question with accurate facts and correct citations. One choice swaps the sources, one states false conclusions, and one answers only a tiny piece.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Reading_PrePost_G4-5W28.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: Nanami's missing pencil case and Nasir's look-alike case	Red herring: Nasir's pencil case. It looks just like hers, but his name is inside, and hers turns up in the car.
2	Two facts about monarchs and where each was learned	Answers vary. Check for two facts, each with a source (a book, a website, a teacher, a nature center).
3	Which is a research question?	Where do monarchs spend the winter?
4	Why read more than one source?	Model: One source rarely has the whole answer. Reading several lets a researcher combine facts and check that they agree.
5	One sentence that uses both pond facts and names both sources	Model: The pond freezes in December (Source 1), and ducks leave the pond when it freezes (Source 2), so the ducks probably leave in December.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Which source gave the acres? Have the monarchs fully recovered?	Source C. Not yet: the colonies grew from 2.22 to 4.42 to 7.24 acres, but 7.24 acres is far below the roughly 45 acres of 1996 to 1997, and threats such as milkweed loss, drought and illegal logging remain.
2	Naveen: "Monarchs lay their eggs on oyamel fir trees in Mexico (Source A)."	Model: Source A never says that. Source B explains that the super generation does not lay eggs during diapause and that females lay eggs only on milkweed after they leave Mexico in March. Fixed: In March, female monarchs fly north and lay their eggs only on milkweed (Source B).
3	"About eight months" versus "8 to 9 months"	Model: A careful researcher reports a range, such as about eight to nine months, and cites both sources, because both are reliable and hiding one would not be fair to readers.

#	Prompt	Model answer / grading note
4	Be the editor	a) According to Source B, the super generation lives longer than its parents. b) There were fewer butterflies.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Why eastern monarchs fly to Michoacán and how they are doing; cite two sources	Model: Late-summer monarchs stop breeding and save their energy (Source B), so they can spend the winter in cool, steady oyamel fir forests in central Mexico (Source A). Their colonies grew to 7.24 acres in 2025 to 2026, but that is still far below the 1990s, and threats remain (Source C).